

A Proactive Triple-Mutation MTD Architecture Enhancing AI-Driven Intrusion Detection Against Intelligent Adversaries in CIoT Networks

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Abstract—With the widespread deployment of intelligent consumer-electronics devices and the rapid expansion of consumer Internet-of-Things (CIoT) ecosystems that support next-generation intelligent systems—including smart homes, edge-AI gateways, and wearable networks—the volume and heterogeneity of device-generated network traffic have surged. To mitigate the limitations of traditional static AI-driven network intrusion detection systems (NIDS), proactive defence paradigms such as Network Moving Target Defence (NMTD) have been incorporated into device-edge-cloud architectures. Unfortunately, sophisticated adversaries may infer defence strategies and circumvent these mechanisms—leading to what we term the drug resistance of intelligent adversarial attacks. To counter this, we propose a reinforcement learning-based proactive Triple-Mutation Moving Target Defence (TM-MTD) framework that simultaneously mutates access proxies, IP addresses, and security strategies. The lightweight edge-level salience optimization ensures accurate detection in highly dynamic traffic environments, while the global policy differential optimization strengthens the defense system’s long-term resilience. Experimental results demonstrate that TM-MTD significantly enhances security performance against a variety of intelligent attack models, providing a robust defense framework for next-generation intelligent consumer-electronics systems.

Index Terms—Intrusion detection system, intelligent attack, moving target defense, deep reinforcement learning, endogenous security.

I. INTRODUCTION

THE proliferation of intelligent consumer-electronics devices, coupled with the rapid expansion of CIoT

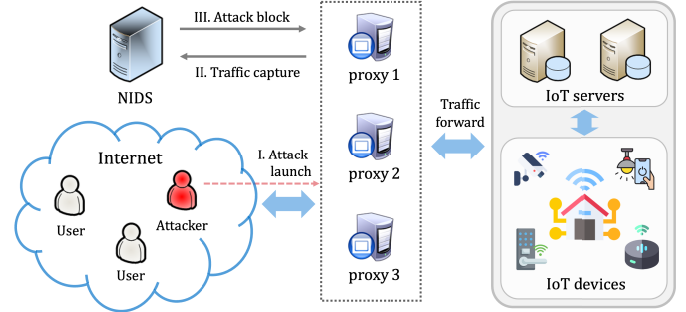


Fig. 1. Overall architecture of network with NIDS in CIoT environments.

ecosystems underpinning next-generation intelligent systems—such as smart homes, edge-AI gateways and wearable networks—has sharply increased both the scale and diversity of network traffic. Meanwhile, numerous industry reports indicate that CIoT environments are now subject to increasingly complex and frequent attack campaigns, including botnets formed by compromised devices, advanced malware targeting embedded consumer systems, and newer forms of intelligent adversarial attacks [1]. According to a 2025 report by cloud-native security firm Zscaler, IoT-based malware attacks against government systems surged by 370% in the past year, while mobile threats grew by 147% [2]. In response to emerging network attack threats, AI-based NIDS solutions have experienced rapid advancements, showcasing robust generalization capabilities, exceptional detection accuracy, and high efficiency [3]. Figure 1 illustrates the overall architecture of the NIDS within the CIoT environment. It can be observed that the NIDS performs intrusion detection by monitoring and analyzing traffic destined for the IoT network, thereby enabling the identification and blocking of malicious traffic before it reaches the IoT environment.

A. Challenges From Intelligent Adversarial Attacks

Despite the increasing use of advanced AI-driven NIDS in CIoT ecosystems [4], [5], these systems have also led to the rise of more sophisticated intelligent adversarial threats. Specifically, intelligent adversarial attacks introduce a “resistance” phenomenon, similar to drug resistance in biology. When facing AI-driven NIDS, attackers continuously analyze and adapt to the defense strategies, refining their methods to bypass detection. Surveys on IoT-related adversarial machine-learning vulnerabilities reveal that these attacks are growing in complexity, yet there is still a significant need for more effective solutions in CIoT settings [6].

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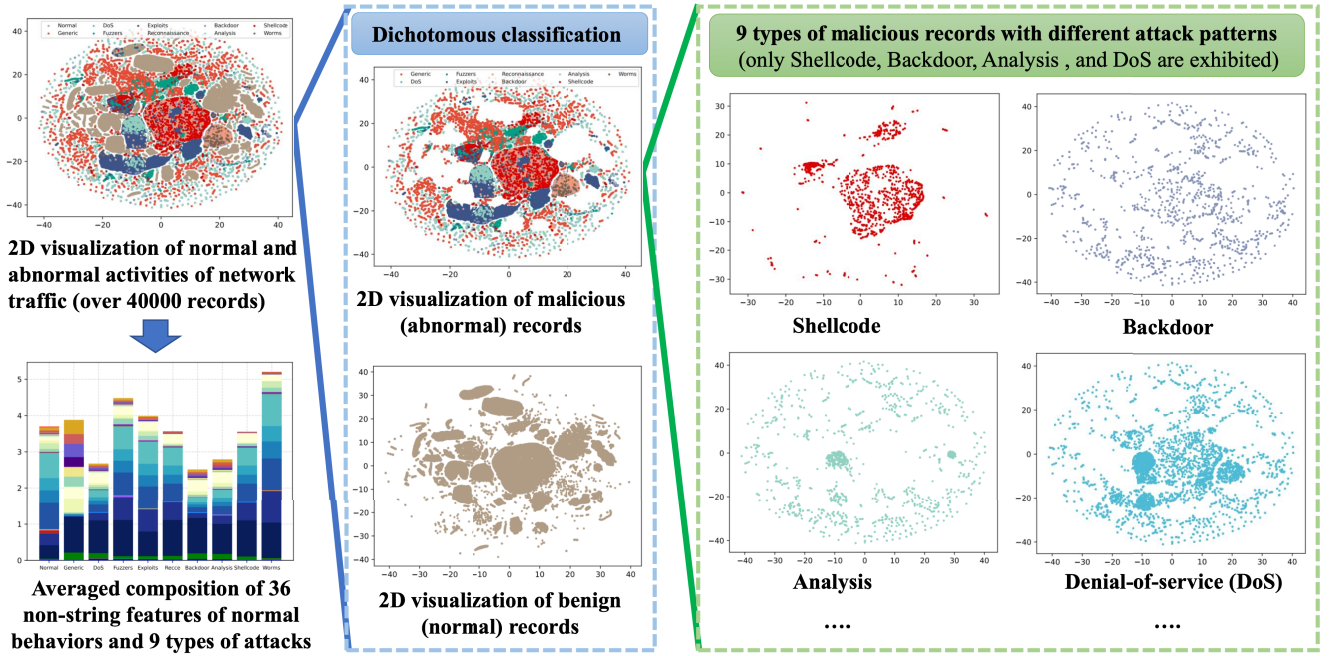


Fig. 2. The feature distributions and characteristic analysis of records from UNSW-NB15 [8], [9]. UNSW-NB15 is a public network intrusion detection dataset with 10 types of malicious/benign network traffic records. The 2D visualization and bar graph on the left-hand side are depicted based on over 40,000 (out of 2 million) randomly sampled records with 36 features. On the right-hand side, Here we only exhibit 4 types of attacks for simplicity.

In order to provide a lucid demonstration of the viability of intelligent attacks, we perform a straightforward analysis using the well-established intrusion detection UNSW-NB15 dataset [7], as illustrated in Figure 2. It can be observed that, due to diverse attack paradigms, such as Shellcode and DoS attacks, malicious records exhibit considerable heterogeneity among themselves, thereby providing ample opportunity for intelligent adversaries to exploit. Attackers can adjust malicious traffic to circumvent defensive strategies.

B. Contributions

To address the limitations of existing NIDS technologies when facing advanced intelligent adversaries, we propose a novel TM-MTD approach that simultaneously mutates access routes, IP addresses, and defense strategies. This method reduces the ability of intelligent attackers to infer defense strategies by reinforcing the defense tactics themselves. As traffic from smart appliances and consumer IoT devices typically passes through multiple proxy or edge nodes before reaching backend services, our method introduces mutations along this routing path, adding an extra layer of uncertainty and disrupting long-term adversarial inference.

Specifically, this work offers contributions in three main aspects:

- First, we systematically analyze the authentic threat posed by intelligent adversarial attacks to prevalent traffic-oriented defense mechanisms. Subsequently, we formulate the endogenous security criteria and establish a hierarchical model for our NMTD framework.
- Next, employing the binary bionic Harris hawks optimization (HHO) algorithm as a foundation, we introduce a

streamlined edge-level adaptation mechanism. This mechanism efficiently determines the alternative feature set for the defense strategy of each edge proxy node, ensuring commendable detection accuracy against a spectrum of traffic attacks.

- Lastly, we introduce a reinforcement learning (RL)-based mechanism to maximize differentiation in the deployment of defense strategies. This entails the development of a novel actor-critic (AC) proximal policy optimization (PPO) framework featuring multiple sub-actors to address the challenge of dimension expansion in detecting abnormal traffic in more diverse CIIoT network.

The paper is structured as follows: We commence with a systematic review of related works in Section II. Then, the threat model and framework of the proposed TM-MTD solution are briefly illustrated in Section III and IV, respectively. Following that, Section V formulates the problem. In Section VI, we provide a comprehensive depiction of our hierarchical active defense solution. Section VII presents the experimental results using representative datasets. Section VIII discuss the limitations and challenges of deploying TM-MTD. Lastly, we conclude and outline potential future directions in Section IX.

II. RELATED WORKS

Active defenses such as NMTD are introduced to proactively disrupt the initial phase of information gathering, thereby enhancing protection against traffic-centric attacks in IoT environments [10]. NMTD defenders strategically alter the attack surfaces of target systems, rendering attacks theoretically ineffective [11]. As illustrated in Figure 3, existing NMTD schemes often involve only two tiers of periodic

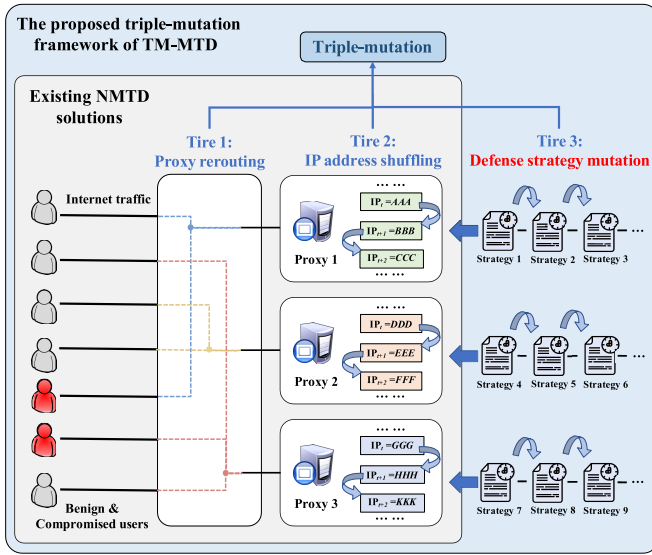


Fig. 3. The core idea of the proposed triple-mutation NMTD framework.

dynamic mutation, namely route redirection between users and proxies and IP address hopping [12]. While effective against conventional attack methods such as stealthy infiltration and eavesdropping harvesting, these strategies have limitations. Thus, solutions like [13], [14], [15] that employ fixed, singular defense strategies are inadequate.

Given that the interaction between attackers and defenders inherently aligns with the principles of game theory, numerous game-based NMTDs have been developed [16], [17]. Reference [18] presents a Markov game-based defense timing strategy for moving target defense in satellite computing systems, improving security efficiency under resource and traffic constraints. Reference [19] develops a four-stage attack-defense game model that yields Nash-equilibrium secure control strategies to improve system resilience. However, as intelligent adversarial attackers deepen their understanding of defense systems, the generated adversarial samples evolve through iterative learning, making it extremely challenging and impractical to clearly characterize attack strategies in advance. Consequently, prior knowledge-supported methods such as game theory [20], [21] are not applicable.

Recent advancements in artificial intelligence have significantly enhanced the ability to counter intelligent attacks [22]. AI has been increasingly utilized to improve the performance of NIDS by boosting detection accuracy and efficiency. For instance, [23] presents a knowledge distillation-based framework for NIDS that achieves similar performance with reduced complexity, size, and inference time. Similarly, [24] introduces a method for smart-home network intrusion detection using graph attention and SHAP for feature selection.

AI also strengthens the adaptability and intelligence of moving target defense architectures [25]. For example, [26] proposes a reinforcement learning-based framework to enhance the resilience of defense agents against adversarial attacks, while [27] uses reinforcement learning with behavioral fingerprinting to guide the selection of MTD techniques against heterogeneous malware on single-board devices.

While existing defense mechanisms have made significant strides in addressing intelligent attacks, they often focus solely on increasing the complexity of intrusion detection strategies. However, these approaches still face challenges in dynamic and resource-constrained CIoT environments, where high resource demands and the need for frequent system updates limit their applicability. In contrast, the proposed TM-MTD framework shifts the focus to enhancing dynamic heterogeneity in defense strategies, incorporating simultaneous mutations of access proxies, IP addresses, and security policies. This approach prevents attackers from converging on a single defense strategy, ensuring long-term resilience.

III. THREAT MODEL

In this paper, therefore, instead of introducing more complex security technologies or defense architectures, we strengthen the dynamism of the NMTD defense strategy so that intelligent adversarial attacks, which rely on reasoning and adaptive countermeasures, are substantially hindered in inferring the underlying defensive mechanisms. Intelligent attackers leverage AI technologies, such as deep learning and reinforcement learning, to enhance their inferential abilities, generating high-precision attack samples that can bypass defense systems through prolonged attack-defense interactions.

A. Signals for Inference

Intelligent attackers rely on observable feedback signals to infer defense strategies, including binary detection feedback, confidence score variations, attack success rates, and packet loss patterns. These signals reveal the defender's actions and allow attackers to refine their strategies. For example, attackers can track attack success rates over time to identify system vulnerabilities.

B. Objective

Attackers aim to finely adjust the generated attack sample x^{adv} (i.e., malicious traffic data) based on the original attack sample x^{org} through AI means to maximize the likelihood of NIDS misclassifying malicious data, ultimately bypassing the defense system.

C. Capability

Unlike adversarial examples in the field of computer vision, network traffic features often have specific value ranges due to the constraints of real network environments. Otherwise, the attack may be invalid, which aligns with common knowledge. Hence, we can depict this constraint as $\|x^{org} - x^{adv}\|_p \leq \epsilon$, where ϵ denotes the upper bound of deviation between original and adversarial samples. The deviation is quantified by p -norm.

D. Attacker Cost and Budget

The attacker iteratively refines adversarial samples based on feedback signals, adjusting the attack strategy to bypass the defense system. The cost of this iterative process is directly tied to the complexity and adaptability of the defense. As

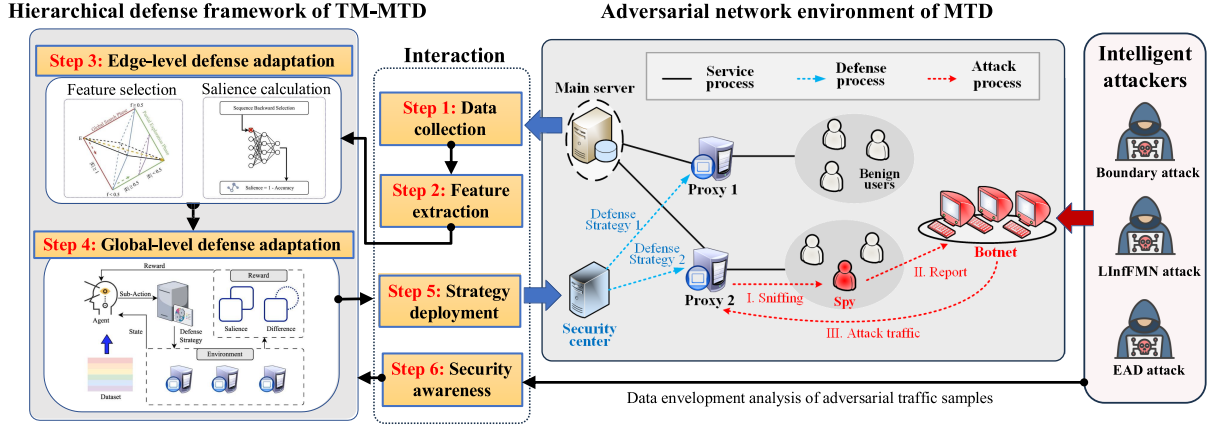


Fig. 4. The structure and offense-defense confrontation of TM-MTD.

defenses become more dynamic and difficult to predict, the attacker is forced to explore a larger space of potential configurations, resulting in higher computational costs and longer convergence times. Thus, the effectiveness of the defense system not only depends on its detection capabilities but also on how resource-intensive it makes the attacker's adaptation process.

Based on the analysis presented above, we can define the threat model of intelligent adversarial attacks addressed in this paper as follows:

$$\begin{aligned} \tilde{x}^{adv} &= \min_{x^{adv}} \mathcal{L}(f(x^{adv}), y) \\ \text{s.t. } \|x^{org} - x^{adv}\|_p &\leq \epsilon, \end{aligned} \quad (1)$$

where y denotes the target label. $f(\cdot)$ represents the attackers' model. $\mathcal{L}(\cdot)$ is the loss function of victim NIDS.

IV. THE OVERVIEW OF PROPOSED METHOD

In response to the threats posed by intelligent adversarial attacks, we propose an innovative secure framework known as TM-MTD. The system leverages a hierarchical architecture, as depicted on the left-hand side of Figure 4. The core concept entails initiating defense adaptation at the edge level to select optimal independent abstract features for constructing the mutation space of MTD. Subsequently, based on cross-domain defense adaptation across the entire network, it generates a defense strategy characterized by substantial inner heterogeneity and robust detection performance. Consequently, when confronted with multiple intelligent adversarial paradigms [28], [29], [30], the defense system is adept at confusing sophisticated adversaries to the maximum extent through shuffling, thereby impeding adversarial traffic samples from bypassing defense measures through adversarial training. This process encompasses the following sequential steps:

Step 1 (Data collection): Both flow-based and packet-based information of traffic are collected from each proxy.

Step 2 (Feature extraction): Quantify and normalize the features of network traffic to create NIDS feature matrices.

Step 3 (Edge-level defense adaptation): The defense system conducts single-point feature optimization, further processing the initial feature set to form independent abstract

features. Based on HHO algorithm, it leverages DL models (e.g., MLP, CNN, LSTM) to select advantageous features and construct the mutation space of NMTD.

Step 4 (Global-level defense adaptation): Subsequently, to enhance the system's endogenous security, unlike existing NMTDs focusing solely on single-point strategy optimization, the defenders engage in network-wide differentiated strategy generation. While ensuring excellent overall defensive performance, TM-MTD maximizes the diversity of defense strategies to confuse intelligent adversaries, preventing the convergence of adversarial sample models.

Step 5 (Strategy deployment): The heterogeneous defense strategies are strategically allocated to each proxy to counter intelligent deduction while maintaining robust and effective performance during periodic shuffling.

Step 6 (Security awareness): Furthermore, we continuously monitor the network security posture and establish a trigger threshold θ . If the situation changes beyond the threshold (i.e., $Thr > \theta$), the defense system will be triggered and dynamically reconfigured, ensuring the adaptability and resilience of the defense mechanism.

The TM-MTD mechanism effectively disrupts the consistency of feedback signals available to attackers by introducing heterogeneous defense strategies, temporal instability, and inter-proxy variability. Specifically, dynamic mutations of access proxies, IP addresses, and defense strategies prevent attackers from building stable models of the defense system, hindering their ability to predict defensive actions and adjust attack strategies accordingly. The diversity of defense strategies ensures attackers cannot generalize from one interaction to the next, making it difficult for attackers to exploit consistent patterns in the defense behavior.

V. PROBLEM FORMULATION

This section provides a concise clarification of TM-MTD's system model. Next, we formulate the problem based on the Markov decision process (MDP) [31]. Consequently, the task of selecting the optimal feature composition becomes synonymous with finding the optimal policy within this MDP. Since our proposal incorporates both malicious traffic detection and endogenous security into its considerations,

we enhance the accuracy of evaluating feature composition (defense strategy) by incorporating augmented endogenous security criteria when defining the reward function.

Figure 4 illustrates the fundamental structure and confrontation process of NMTD against malicious traffic initiated by sophisticated attackers. In this setup, where the network system channels service flows through multiple edge proxy servers, adversaries begin by infiltrating spies (compromised users or forged requests) to infiltrate and gather information from the targeted proxies. Subsequently, these spies relay the collected information, enabling the Botnet to weaponize the malicious traffic and initiate the attack. Conversely, NMTD defenders periodically reassign users among different proxies while assessing the accumulated malice of each user over time. This efficient strategy leads to the elimination of stealthy spies after several rounds of interaction, rapidly depleting the attackers' resources and ultimately terminating the attacks within a short timeframe.

Leveraging the proposed active defense architecture, which prioritizes endogenous security enhancement while addressing salience and variance objectives, we can cast the NMTD policy optimization problem following MDP. We define a five-tuple $\langle \mathbb{S}, \mathbb{A}, P, R, \gamma \rangle$ to represent this formulation: $\mathbb{S}$ signifies the state space, $\mathbb{A}$ represents the action space, P denotes the state transfer matrix, which characterizes the likelihood of the current state $S^t \in \mathbb{S}$ transitioning to the subsequent state S^{t+1} after the adoption of action $A^t \in \mathbb{A}$ at round t ; $R(S^t, A^t)$ denotes the reward function associated with the state-action pair (S^t, A^t) ; and γ serves as a time discount factor, balancing future and current rewards within the decision-making process.

A. State Space

To enable the mutation of defense strategies, time is divided into slots represented by $t \in [1, T]$. Each t represents the smallest unit of time during which TM-MTD evaluates policy earnings based on the current state of the defense system and transitions to the next slot's defense strategy.

The current state comprises three key components: the adjacency matrix D^t , the global defense strategy deployment F^t , and the detection importance W^t . This state is represented as $S^t = (D^t, F^t, W^t)$.

The first term, denoted as $D^t = \{d_{i,i'}^t | \forall i, i' \in N\}$, represents the connection paths between proxy nodes. Here, 1 signifies availability, while 0 signifies unavailability. Given the relative stability of these connection relations despite different actions, we simplify the analysis in subsequent sections by setting $d_{i,i'}^t = 1$ for all i and i' in N , and t in T .

The second term, denoted as $F^t = \{f_{i,j}^t | \forall i \in N, j \in M\}$, represents the present defense strategy, encompassing feature selections for all proxies within the system at time slot t . Specifically, $f_{i,j}^t = 1$ indicates that proxy i has chosen the j^{th} feature, while $f_{i,j}^t = 0$ signifies the opposite.

The third term, denoted as $W^t = \{w_{i,j}^t | \forall i \in N, j \in M\}$, signifies the importance of features in detecting malicious traffic. Each $w_{i,j}^t$ is a time-varying value within the range $[0, 1]$. When $w_{i,j}^t$ is close to 1, it indicates that feature j holds significant importance as a detection dimension. Conversely, when $w_{i,j}^t$ deviates from 1, feature j becomes less relevant for adversary identification.

Additionally, taking into account the constrained sensing and computational capabilities of network edge nodes in real-world scenarios, we impose restrictions on the number of features that a single proxy can select as follows:

$$\sum_{j=1}^M f_{i,j}^t \leq C_i, \quad \forall i \in N \quad (2)$$

where M is the number of features in the optional feature set and N denotes the number of proxy nodes.

B. Action Space

For each edge proxy server, the defender can adjust the defense strategy F_i^t by adding, preserving, or removing one or more features during time slot t . We denote these operations as atomic actions 1, 0, -1, respectively. Consequently, the TM-MTD action is represented as a multidimensional boolean composite vector $A^t = A_1^t, \dots, A_N^t$ for all $t \in T$. Here, $A_i^t = a_{i,1}^t, \dots, a_{i,M}^t$ represents the malicious traffic detection formula for proxy i , with each $a_{i,j}^t$, where $1 \leq i \leq N$ and $1 \leq j \leq M$, constituting an atomic action.

We denote the action space as $\mathbb{A}$. Its size grows exponentially with the number of edge proxy servers and features, reaching $3^{N \times M}$, which is infeasible for large-scale network traffic protection. Moreover, edge-side computational constraints and state conflicts introduce numerous illicit actions. For instance, $a_{i,j}^t = 1$ is invalid when $f_{i,j}^t = 1$, as a feature cannot be selected twice. Section VI addresses these issues by removing illicit actions and reducing action-space complexity to ensure computational feasibility.

C. State Transition

The alterations in the feature composition of proxies are construed as state transitions within the defense framework. Given that the state at the next time slot, denoted as S^{t+1} , solely depends on the current state, S^t , and the action taken, A^t , this transition process possesses the Markov property. Consequently, we represent the probability of this state transition as $Pr(S^{t+1}, R^{t+1} | S^t, A^t)$. To illustrate, consider a scenario with a single-proxy system encompassing three feature dimensions, where the current strategy deployment is $F^t = 0, 1, 1$. In the event that action $A^t = 1, 0, -1$ is executed, the resulting strategy deployment becomes $F^{t+1} = F^t + A^t = 1, 1, 0$. This sequence is succinctly denoted as $S^t \xrightarrow{A^t} S^{t+1}, R^{t+1}$, encapsulating the state transition and the associated reward.

Nonetheless, despite the deterministic nature of the transition rules governing F^t , the presence of the dynamic variable W^t renders the state transition of TM-MTD's S^t inherently non-deterministic. Consequently, in the subsequent section, we propose the utilization of a Reinforcement Learning (RL)-based mechanism to address this inherent uncertainty.

D. Endogenous Security-Involved Reward Function

The reward function represents the security profit of defenders with a particular state-action pair. Maximizing the average reward for the whole process is the ultimate goal of optimization problems. As TM-MTD jointly requires high detection

accuracy against malicious traffic and endogenous security of the defense system itself, we define the reward function $R_\pi(S^t, A^t)$ as the sum of total strategy salience and global strategy difference, which can be denoted as $G_g(S^t)$ and $G_d(S^t)$, respectively.

Since the features are not completely independent from each other, we decoupled the correlation between the features of the original dataset using Independent Component Analysis (ICA) before calculating the salience of each feature.

The global difference is calculated using the Pearson Correlation Coefficient of the proxy nodes' deployment features. Pearson Correlation measures the linear relationship between two variables, ranging from -1 (perfect negative correlation) to 1 (perfect positive correlation), with 0 indicating no correlation. Since we aim to maximize the uncorrelation between defense strategies, the goal is to minimize this coefficient. The details are as follows:

$$G_d(S^t) = 1 - \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{j=-i}^N |\rho(s_i^t * w_{i,j}^t, s_j^t * w_{j,i}^t)| \quad (3)$$

where $\rho(X, Y)$ is the Pearson correlation coefficient between X and Y .

The strategy salience index, denoted as $G_g(S^t)$, quantifies the cumulative salience of the deployed features. It is defined as follows:

$$G_g(S^t) = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M s_{i,j}^t \cdot w_{i,j}^t, \quad (4)$$

where $w_{i,j}^t$ signifies the salience (detection accuracy) of the j^{th} feature on the i^{th} proxy at round t , with values in the range of $[0, 1]$. A higher value of $G_g(S^t)$ indicates that the defense strategies of proxy servers are more precise and effective in eliminating malicious spies.

Hence, the ultimate composite reward can be formulated through the weighted synthesis of equations (3) and (4), expressed as:

$$R^t(S^t, A^t) = (1 - \lambda) \cdot G_g^t + \lambda \cdot G_d^t - \tilde{R}^{br} \quad (5)$$

where $\lambda \in [0, 1]$ serves as an adaptable weight parameter that harmonizes the objectives of endogenous security enhancement and detection performance optimization. $\tilde{R}^{br}$ is the base reward. Subtracting base reward is one of the common practices in reinforcement learning that can improve the stability and effectiveness of training. If the reward is kept positive, it may lead to an over-concentration of the reward signal, making the training unstable. In our setup, base reward is defined as the top 20% of the randomly generated strategies.

E. Optimization Goals

The primary objective of the defense strategy is to enhance the accuracy and efficiency of detection and elimination of malicious traffic. In essence, the selected features should maximize the evaluation scores of attacker malice while minimizing those of normal users. Here, we utilize $\pi(A^t|S^t)$ to denote the probability of selecting an action A^t (i.e., a defense strategy) in a given state S^t . The secondary optimization goal is to maximize the overall disparity among defense strategies across

all proxies in the system, aimed at countering the impact of intelligent attacks.

$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_{t \in T} \zeta^t R(S^t, A^t) \mid S^0 = s, A^t \right] \quad (6)$$

where $A^t \sim \pi(\cdot | S^t)$, ζ represents the discount factor. It is important to note that ζ serves a different purpose compared to the previously discussed weighting parameter λ , as it controls the emphasis on future rewards in the optimization process, while λ governs the trade-off between different objectives in the defense strategy.

F. Problem Formulation

Considering the dual security perspectives inherent in TM-MTD, the optimization problem aimed at finding the optimal solution for defense strategy deployment can be formally defined as:

$$\begin{aligned} \text{Maximize : } & \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \zeta^t R(S^{t+\tau}, A_{\pi}^{\tau}) \mid S^{\tau} \in \mathbb{S} \right] \\ \text{Subject to : } & \sum_{j=1}^M f_{i,j}^t \leq C_i, \quad \forall i \in N \\ & f_{i,j}^t \in (0, 1), \quad \forall i \in N, \quad j \in M \end{aligned} \quad (7)$$

VI. HIERARCHICAL DEFENSE FRAMEWORK

In this section, we provide a comprehensive exposition of the TM-MTD framework. The proposed hierarchical active defense framework comprises two key layers: an edge-level feature selection layer, which focuses on optimizing detection salience at the individual edge proxy nodes, and a global strategy adaptation layer, which seeks to bolster the system's inherent security by maximizing strategy disparities among the proxy nodes.

A. Edge-Level Defense Adaptation

The growing complexity of network services and advances in network-awareness technologies have dramatically expanded traffic feature dimensions, many of which are irrelevant or ineffective for detection. Thus, an effective defense strategy must begin with a suitable feature extraction or selection mechanism.

In the CIoT environment, each proxy node must adaptively select an effective set of traffic features under limited computational resources and in response to dynamically evolving attack behaviors. This presents a high-dimensional, non-convex feature selection problem, which is well-suited to HHO due to its strong exploration capability, fast convergence, and low computational overhead. Compared to Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and traditional greedy methods, HHO offers significant advantages. It converges more quickly by balancing exploration and exploitation effectively, while also reducing computational complexity. In contrast, GA and PSO are prone to slower convergence and local minima, and greedy methods lack the global search capability needed for high-dimensional problems.

In our system, each HHO agent represents a candidate defense configuration, i.e., a specific subset of traffic features activated for intrusion detection at an edge proxy. The position of an agent corresponds to a concrete feature-selection strategy, and the population collectively explores alternative configurations to avoid easily inferable defenses. The exploration–exploitation mechanism of HHO maps naturally to defensive behaviors: global exploration encourages proactive diversification of feature sets to resist intelligent adversarial probing, while local exploitation refines high-performing configurations that exhibit strong detection salience.

We have orchestrated the deployment of a cohort of Q agents, each responsible for periodic updates to their respective solutions denoted as $X_q^t, q \in Q$, working collectively towards the attainment of the optimal outcome. To commence this collaborative effort, we initialize two critical parameters: the escape energy E and the escape probability f . The energy term E characterizes the intensity of defense strategy adjustment, that is, the tendency of an edge node to actively explore alternative feature-selection configurations rather than merely refining its current strategy. Notably, f is a stochastic variable that assumes values within the range of 0 to 1. Over the course of iterations, the escape energy E undergoes dynamic adjustments, as defined by the equation:

$$E = 2E_0 \left(1 - \frac{t}{T}\right), \quad (8)$$

where t signifies the current time slot, T corresponds to the maximum iteration rounds, and $E_0 \in [-1, 1]$ stands for the initial energy value, where specifically, $E_0 = 2f - 1$. Here, t refers to the time slot in the HHO iteration process for selecting the most optimal feature set, which has a different meaning from the t used to represent the number of rounds in the MDP model discussed earlier.

1) *Global Search and Partial Exploration Phases*: Based on the varying values of E , the HHO algorithm operates through the Global Search and Partial Exploration phases. The details of these two phases, including the associated mathematical formulations, are provided in Appendix A (see Supplementary Material). These equations describe how the solutions X_q^t of each agent are updated to X_q^{t+1} in the HHO algorithm.

2) *Edge-Level Defense Adaptation (EDA) Algorithm*: Based on the final outcome achieved after T iterations, defenders can assess the significance of each feature and choose those feature dimensions where the result equals 1. Consequently, a candidate feature set is established for formulating the maliciousness elimination formula through a novel edge-level salience optimization algorithm denoted as EDA. The pseudo-code for EDA is presented in Algorithm 1.

To determine the significance of each feature dimension for the edge server, EDA employs a metric called salience denoted as W , which is calculated as $W = 1 - acc$, where acc represents the accuracy obtained after removing specific feature during the training process. Subsequently, utilizing the calculated detection importance for each feature, a subset of features is generated to serve as the candidate defense strategy space for the proxy.

Algorithm 1 Edge-Level Defense Adaptation (EDA)

Input: number of agents Q , number of iterations T
Output: Selected features M^* , feature salience $W_{M^*}^t$

- 1 Initialize Q agents with random feature composition $X_q^t, 1 \leq q \leq Q$;
- 2 **for** $t = 1 \rightarrow T$ **do**
- 3 **for** $q \in Q$ **do**
- 4 $X_q \rightarrow M_{temp}$;
- 5 Calculate the fitness $F(M_{temp})$ with Eq. (23) in Appendix A and determine the optimal X^* as the solution;
- 6 Update E and J ;
- 7 Based on the values of E and f , update X_q^{t+1} with Eq. (15), (17), (18), (22), and (24) in Appendix A;
- 8 **end**
- 9 **end**
- 10 $X^* \rightarrow M^*$;
- 11 **for each** $m \in M^*$ **do**
- 12 $H = M^* \cup \neg m$;
- 13 Train model with the features in H as inputs;
- 14 Generate the feature salience $w_{i,m}^t = 1 - acc$;
- 15 **end**
- 16 **return** $M^*, W_{M^*}^t$

B. Endogenous Difference Maximization Mechanism of Global Strategy Deployment

In this section, we introduce an RL-based mechanism aimed at enhancing global endogenous security throughout the deployment of defense strategies. However, the issue of dimensionality, often referred to as the curse of dimensionality, presents a significant challenge, making it impractical to directly employ established RL methods such as PPO in the context of NMTDs. To address the challenge posed by the rapidly growing complexity in feature selection, we have devised two distinct modules with specific responsibilities: one for reducing the action space and the other for facilitating action selection.

1) *Sparse and Illicit Problems of Action*: Building upon the MDP model elucidated in Section V, it is theoretically plausible to employ the widely adopted PPO algorithm for the optimization of defense strategy deployment. However, two critical issues have emerged from this approach.

Firstly, each feature associated with an edge proxy entails three atomic actions, represented as $a_{i,j} \in -1, 0, 1$. This results in an action space whose size is exponentially related to the product of the edge proxy servers and the total number of features, potentially reaching as high as $3^{N \times M}$. Consequently, model training consumes a substantial amount of computational resources and time.

Secondly, a significant number of illicit actions have been identified. To illustrate, consider a scenario with three features on a proxy. If the state at time t is $F^t = 0, 0, 1$, with a constraint of $C_1 = 1$ allowing up to one feature to be deployed on this edge proxy, there exist illicit actions such as $A_1^t = -1, 0, 0$ and $A_2^t = 1, 0, 0$. The former erroneously attempts to remove the non-existent 1st feature, while the latter

violates the constraint by transforming F^t to $F^{t+1} = 1, 0, 1$. These illicit actions consume redundant resources, impeding model training efficiency. Consequently, a direct application of the PPO algorithm is deemed impractical.

2) *Action Space Reduction Module*: In our system model (as elaborated in Section V), it's noteworthy that each action is a composite of individual sub-actions denoted as $a_{i,j}^t$. Consequently, our approach deviates from the selection of actions from the set $\mathbb{A}$; instead, we form a comprehensive action by aggregating these pertinent sub-actions.

Additionally, upon reexamining the reward formulation, we observe that the reward is not determined by the specific action type (i.e., addition, retention, or removal), but rather by the system states before and after action execution, as discussed in [32]. Accordingly, we redefine the action representation such that an action $a_{i,j}^t$ indicates whether feature j is deployed on edge proxy server i ($a_{i,j}^t = 1$) or not ($a_{i,j}^t = 0$). This reformulation substantially reduces the dimensionality of the action space from $3^{N \times M}$ to $N \times M$, while completely eliminating the first category of invalid actions, thereby improving both tractability and training stability.

3) *Sub-Action Selection Module*: To address the second challenge related to invalid actions, we design a sub-action selection module that generates a feasible set of sub-actions while explicitly respecting the computational resource constraint C_i at each edge proxy. This module employs neural networks to output the probability distributions over all candidate sub-actions associated with each proxy.

It is worth noting that the reduction of the action space fundamentally changes the action representation. Specifically, at each time step t , the action is no longer a single atomic decision but a set of sub-actions, to which the system reward R^t is jointly assigned. This structural change necessitates corresponding modifications to both action selection and parameter update mechanisms in the PPO framework.

The first enhancement concerns the action selection strategy. In standard PPO, actions are sampled directly from a policy distribution. In our setting, the proposed sub-action selection module is used to construct a valid sub-action set that satisfies resource constraints, generating an optimal and executable defense configuration for each edge proxy.

The second enhancement focuses on parameter updates. Although the executed action A^t is a sub-action set, the policy parameters are associated with individual sub-actions $a_{i,j}^t$. In conventional PPO, the transition tuple $(S^t, A^t, p(A^t | S^t), V_\mu(S^t), R^t)$ is stored for policy and value updates, where $V_\mu(S^t)$ denotes the Critic output. However, the reward R^t assigned to the entire set A^t cannot be directly used to update each constituent sub-action. To resolve this issue, we introduce a scaling mechanism that allocates rewards $r_{i,j}^t$ and value estimates $v_{i,j}^t$ to individual sub-actions in proportion to their selection probabilities. The exact formulations of this scaling process are presented below.

$$\begin{aligned} r_{i,j}^t &= \frac{p(a_{i,j}^t | S^t)}{\sum_{a \in A^t} p(a | S^t)} R^t \\ v_{i,j}^t &= \frac{p(a_{i,j}^t | S^t)}{\sum_{a \in A^t} p(a | S^t)} V_\mu(S^t) \end{aligned} \quad (9)$$

Consequently, the updates for all $a_{i,j}^t \in A^t$ are consolidated. To clarify, the GDA involves multiple updates for a sequence of sub-actions within an episode, rather than performing individual updates for each action.

Next, we are tasked with determining the sub-action composition—namely, the features—for each edge proxy. The process begins by transforming the one-dimensional array of all sub-action selection probabilities generated by the neural network into a two-dimensional array $p(a_{i,j}^t | S^t)$. The selection probability for each feature j on proxy i is normalized through Eq. (10):

$$p(a_{i,j}^t | S^t) = \frac{p(a_{i,j}^t | S^t)}{\sum_{j=0}^M p(a_{i,j}^t | S^t)}, i \in \{1, \dots, N\} \quad (10)$$

Thus, based on the computed probabilities, we proceed to select C_i features from the total M options for the composition of features on proxy i . The culmination of these iterations yields the complete action A^t .

4) *Global Defense Adaptation (GDA) Algorithm*: Subsequently, we amalgamate the aforementioned modules to formulate our multi-update sub-action PPO algorithm aimed at global difference maximization.

The core of GDA lies in Generalized Advantage Estimation (GAE), which more accurately estimates the advantage of a given action relative to the expected outcome. The GAE formula computes the weighted sum of time-difference errors over future steps, allowing the system to incorporate both immediate and future rewards. This helps improve training stability by reducing variance in advantage estimates. GAE is calculated as follows:

$$\hat{H}_t = \delta_t + (\gamma\xi)\delta_{t+1} + (\gamma\xi)^2\delta_{t+2} + \dots + (\gamma\xi)^{B-t-1}\delta_{B-1}, \quad (11)$$

where B signifies the size of the experience replay memory, γ represents the discount factor within the range of 0 to 1, ξ is a pre-defined GAE parameter, and δ_t is the Time-Difference Error (TD-Error), defined as follows:

$$\delta_t = r_{i,j}^t + \gamma v_{i,j}^{t+1} - v_{i,j}^t. \quad (12)$$

δ_t is computed as the difference between the reward at time t and the predicted value of the system's state. This error helps to evaluate how much better or worse the system's current action was compared to the expected action. The goal is to minimize this error, ensuring that the system's actions lead to better long-term outcomes.

The definition of the loss function $\mathcal{J}^{\text{actor}}$ for the Actor network can be expressed as:

$$\mathcal{J}_\theta^{\text{actor}} = \mathbb{E}[\min(r_\theta^t \hat{H}_t, \text{clip}(r_\theta^t, 1 - \epsilon, 1 + \epsilon) \hat{H}_t)], \quad (13)$$

where $r_\theta^t = \frac{p_\theta(a_{i,j}^t | S^t)}{p_{\text{old}}(a_{i,j}^t | S^t)}$, similar to v^t . π_θ is updated using the stochastic gradient descent (SGD) method, where the gradient is the negative of $\nabla \mathcal{J}_\theta^{\text{actor}}$.

The loss function $\mathcal{J}^{\text{critic}}$ for Critic network is defined as:

$$\mathcal{J}^{\text{critic}}(\mu) = \mathbb{E}[(\hat{V}_t^{\text{returns}} - V_\mu(S^t))^2], \quad (14)$$

where the parameters of the target value $\hat{V}_t^{\text{returns}}$ for the TD-Error are determined as $\hat{H}_t + v^t$, and V_μ is updated utilizing the gradient $\nabla \mathcal{J}^{\text{critic}}(\mu)$.

Algorithm 2 Global Defense Adaptation (GDA)

Input: $\mathcal{S}^0$, R^t , number of episodes K , number of epochs T , batch size U

Output: Final state $\mathcal{S}$

- 1 Initialize the weights of both Actor and Critic networks, and the experience replay memory $\mathcal{B}$;
- 2 **for** $k = 1, 2, \dots, K$ **do**
- 3 Reset the environment state as $\mathcal{S}^0$;
- 4 **for** $t = 1, 2, \dots, T$ **do**
- 5 Get action A^t according to current state $\mathcal{S}^t$;
- 6 Calculate sub-action selection probability $p(a_{i,j}^t | \mathcal{S}^t)$, $a_{i,j}^t \in A^t$ and state value $V_\mu(\mathcal{S}^t)$;
- 7 **for** $a_{i,j}^t \in A_t$ **do**
- 8 Calculate $r_{i,j}^t$ and $v_{i,j}^t$ with Eq. (9);
- 9 Collect $\mathcal{D}_{a_{i,j}^t} = \langle \mathcal{S}^t, a_{i,j}^t, p, v_{i,j}^t, r_{i,j}^t \rangle$;
- 10 $\mathcal{B} = \mathcal{B} \cup \mathcal{D}_{a_{i,j}^t}$;
- 11 **end**
- 12 Execute each $a_{i,j}^t \in A_t$ to obtain the new state $\mathcal{S}^{t+1}$ and the reward R^t accordingly;
- 13 **end**
- 14 Calculate the GAE $\hat{H}_t$ with Eq. (11);
- 15 Retrieve B/U mini-batches from $\mathcal{B}$;
- 16 **for every batch do**
- 17 Update Actor network with Eq. (13);
- 18 Update Critic network with Eq. (14);
- 19 **end**
- 20 **end**
- 21 **return** $\mathcal{S}$

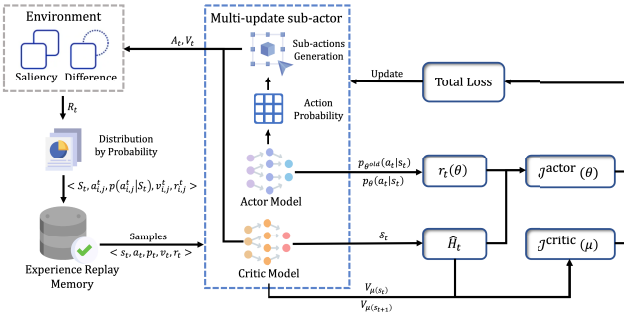


Fig. 5. The architecture of GDA's training process.

The loss function for the Actor network is designed to optimize the policy by encouraging actions that lead to higher rewards while maintaining stability in learning. The Critic network evaluates the state value function, helping to improve the accuracy of the reward predictions. The combination of these two networks enables the system to both select the best actions and accurately predict the outcomes of those actions.

The pseudo-code detailing the proposed global difference maximization algorithm is presented in algorithm 2.

5) *The Architecture of GDA's Training Process:* As depicted in Figure 5, our GDA architecture has been extended to accommodate sub-action selection and iterative updates within the training process. Specifically, we have introduced an additional sub-action generation step into the Actor-Critic-based composite neural networks. Prior to storing samples in

the experience replay memory, a probability distribution step has also been incorporated.

Within each episode k , the initial state is reset to $\mathcal{S}^0$. Subsequently, during each epoch t , we employ the strategy $\pi_{\theta_{old}}$ and the current state $\mathcal{S}^t$ to generate action A^t . This includes determining the selection probability $p(a_{i,j}^t | \mathcal{S}^t)$ for each sub-action $a_{i,j}^t \in A^t$, as well as estimating the state value $V_\mu(\mathcal{S}^t)$ through the Critic network.

After executing each sub-action $a_{i,j}^t \in A^t$, we observe the resultant state $\mathcal{S}^{t+1}$ and the corresponding reward R^t . For each sub-action $a_{i,j}^t \in A^t$, we calculate the reward $r_{i,j}^t$ and the value $v_{i,j}^t$ in accordance with formula (9), while considering the associated probabilities. This information is stored as $\langle \mathcal{S}^t, a_{i,j}^t, p(a_{i,j}^t | \mathcal{S}^t), v_{i,j}^t, r_{i,j}^t \rangle$ in the experience replay memory denoted as $\mathcal{B}$. Subsequently, we perform B random sampling operations from $\mathcal{B}$ to compute the gradients for both the Actor and Critic networks.

VII. PERFORMANCE EVALUATION

In this section, we conduct a comprehensive evaluation of the security performance of our proposed TM-MTD solution. We begin by providing a detailed exposition of the experimental setup. Next, we conduct training convergence and stability experiments, followed by an analysis of the weighting parameter λ selection. We then evaluate the defense performance of our TM-MTD methodology against a series of baselines, followed by an ablation study to validate the contributions of each mutation component. Finally, we perform latency experiments under varying numbers of proxies to assess the scalability of the system.

A. Experimental Environment

To conduct a series of traffic active defense experiments, we construct the testing environment based on a workstation equipped with Intel i9-10940X CPU, 128GB RAM, two RTX3090 24GB GPUs, and Windows 10-21H1 OS. All the programs are realized based on Python 3.9 with PyTorch.

The experimental configuration comprises a primary server, designated as the global control node, and a network of ten accessible proxy servers functioning as edge nodes. Our method's efficacy is evaluated against benchmark intrusion detection datasets rooted in flow-based network traffic: UNSW-NB15 [7] and CICIoT2023 [33]. UNSW-NB15 encompasses an extensive collection of approximately 250,000 network traffic samples, characterized by a feature set of 49 attributes and the inclusion of 10 distinct attack paradigms. In contrast, CICIoT2023 comprises over 680,000 network traffic samples, distinguished by a feature space comprising 46 attributes and a comprehensive coverage of 7 distinct attack categories.

B. Convergence and Training Stability Analysis

After the edge nodes identify satisfactory candidate feature sets $F_i^t, \forall i \in N$, the RL-based GDA algorithm is initialized at the control server and trained for $K = 1000$ episodes. The batch size U is set to 256, the discount factor γ is 0.95, and each update cycle includes eight gradient update iterations.

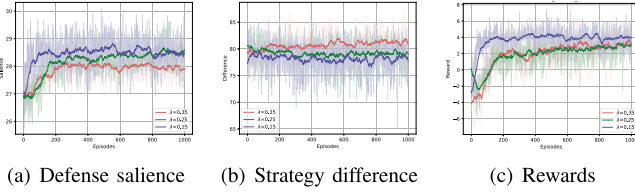


Fig. 6. The training stability of the proposed algorithm.

Both the Actor and Critic networks use learning rates of 0.0003, with the PPO clipping parameter fixed at 0.2. A single hidden layer with 512 neurons is employed for all networks. All hyperparameters are kept identical across experiments to isolate the impact of the adaptive parameter λ .

To examine convergence, training stability, and the sensitivity of the weighting parameter λ , we analyze the evolution of defense salience, strategy difference, and cumulative reward under different values of λ . As illustrated in Figures 6(a) and 6(b), all tested configurations demonstrate stable convergence after an initial exploration phase, indicating that the learning process does not suffer from divergence or oscillation. Among these configurations, $\lambda = 0.15$ leads to faster convergence and achieves higher steady-state salience while maintaining a sufficient level of global strategy differentiation.

As shown in Figure 6(c), the configuration with $\lambda = 0.15$ shows the fastest convergence and the most stable reward plateau, with reduced variance in the later stages of training compared to other values of λ . Overall, these results indicate that the proposed GDA algorithm converges reliably across different parameter settings, with $\lambda = 0.15$ offering an optimal balance between convergence speed, stability, and long-term performance.

C. Evaluation of Defense Performance

To further examine the general defense performance of TM-MTD against different intelligent adversaries, we compare the performance of TM-MTD with other types of NMTD technologies: Dynamite defense [34], randomly selected solution [35] and static optimal solution [36]. The random approach utilizes randomly generated defense strategies alongside agent switching, while the static approach deploys the best defense strategy for each proxy after RL. The Dynamite method uses dynamic strategy adjustment based on adversarial feedback. Besides, we also launch three intelligent adversarial attacks, including both glass box and closed box paradigms. The adopted adversarial attacks include boundary attack [28], EAD attack [30] and LInFMN attack [29].

We evaluated four defense methods—Our Method, Dynamite, Random, and Static—across three different adversarial attack types: Boundary Attack, EAD Attack, and LInFMN Attack under the UNSW-NB15 and CICIOT2023. The attacker's attack success rate (ASR) was used as the key metric, and the results are shown in Figures 7.

The results demonstrate that our proposed method consistently achieves the lowest attack success rate in all six attack-defense configurations. Notably, our method significantly reduced the LInFMN attack success rate to 4.76% under the CICIOT2023 dataset. While a brief fluctuation in

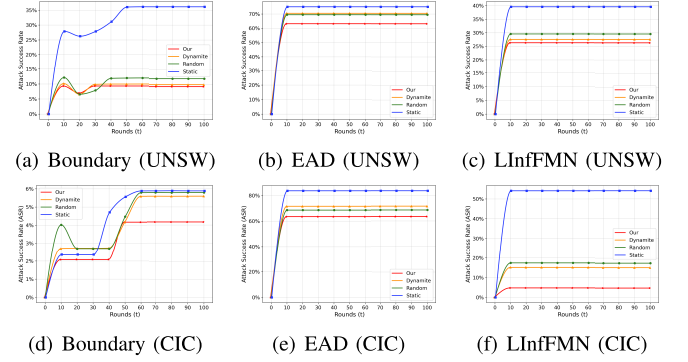


Fig. 7. The attack success rate against different NMTDs.

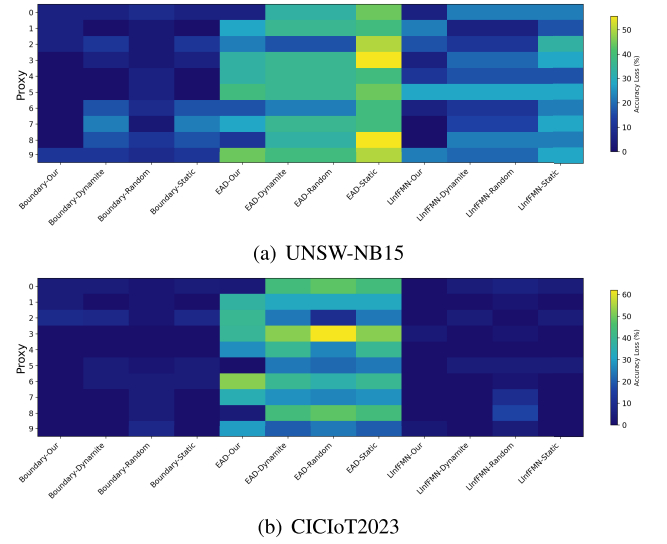


Fig. 8. The proxy-specific accuracy degradations against intelligent attacks.

attack success rate above the Dynamite and Random methods was observed for Boundary Attack against our method, the value remained within an acceptable range.

Among all attack types, EAD attacks proved to be the most effective, achieving an attack success rate of 83.6% against the Static defense method. In contrast, our method reduced the EAD attack success rate to 63.4%, the lowest among the four defense strategies, highlighting the superiority of our approach in handling strong adversarial attacks.

These results underscore the effectiveness of our method in mitigating a wide range of adversarial attacks, and its strong performance even under challenging scenarios with higher-dimensional datasets and more sophisticated attack types.

The heatmaps in Figures 8 continue to depict the varying impact of attacks on different proxy nodes. We define *accuracy loss* as the metric for evaluating defense strategies, calculated by subtracting the accuracy under adversarial attacks from the original accuracy of the NIDS model. Under the Boundary and LInFMN attacks, the four defenses exhibit relatively balanced Accuracy Loss. Notably, the EAD attack results in significantly high accuracy loss for some of the proxy nodes where our strategy is deployed, in some cases even surpassing or closely approaching the losses observed with the other two methods. However, its overall defensive performance remains substantially superior to other approaches. This underscores

TABLE I
RESULTS OF THE ABLATION STUDY

Configuration	ASR (%)	Accuracy Loss (%)
No IP Mutation	72.1	55.2
No Route Mutation	71.4	52.1
No Strategy Mutation	83.3	63.3
Full Three-Mutation Setting	63.3	46.3

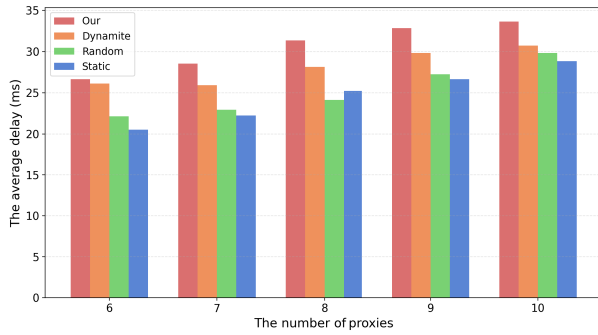


Fig. 9. The average delay under different numbers of proxies.

the strength of our defense, emphasizing that it does not rely solely on the absolute superiority of a single strategy but leverages multiple diverse defense tactics to collectively confound attackers, thereby establishing long-term resilience. Detailed experiments on accuracy loss are presented in Appendix B (see Supplementary Material).

D. Ablation Study on the Impact of Mutation Components

To better understand the individual contributions of each mutation component in our proposed TM-MTD framework, we conducted an ablation study by progressively removing the three mutation components: IP mutation, route mutation, and strategy mutation. The experiments were conducted using the most powerful of the three attacks, the EAD attack. The results of this experiment, including the ASR and accuracy loss, are summarized in Table I.

The ablation study results demonstrate that each mutation component contributes to the overall performance of TM-MTD. Removing the strategy mutation results in the largest increase in both ASR (83.3%) and accuracy loss (63.3%), highlighting the critical role of strategy mutation in preventing attackers from converging on a single defense strategy. The removal of IP and route mutations also leads to increases in ASR and accuracy loss, but to a lesser extent, emphasizing their importance in enhancing defense robustness.

The full three-mutation setting yields the best defense performance with an ASR of 63.3% and an accuracy loss of 46.3%. These results confirm the superiority of our method, with each mutation playing a significant role in balancing attack success rate and accuracy loss. More detailed experimental results can be found in Appendix C (see Supplementary Material).

E. Impact of Proxy Number on Average Delay

We conducted an experiment to evaluate the impact of varying the number of proxies on the average delay for four aforementioned defense methods. The results, illustrated in Figure 9, show that as the number of proxies increases

from 6 to 10, the average delay for all four methods also increases. This increase is expected as the additional proxies introduce more network overhead and require more time for communication and feature processing.

Notably, our method and Dynamite exhibit slightly higher delays compared to the Random and Static methods for the same number of proxies. However, the incremental increase in delay is relatively small, approximately 5 milliseconds, which is within an acceptable range for most real-time systems. This suggests that while our method may introduce some overhead due to its dynamic nature, it remains efficient and practical for CIoT environments, where system responsiveness is crucial.

Furthermore, the experiment demonstrates that the proposed method maintains reasonable scalability with respect to the number of proxies. The increase in delay remains moderate even as the number of proxies grows, indicating that our method can handle larger-scale deployments effectively. This scalability is crucial for ensuring that the defense mechanism can be deployed in diverse CIoT settings with varying numbers of edge proxies, without compromising system performance to an unacceptable extent.

VIII. LIMITATIONS

While the proposed TM-MTD framework offers enhanced protection against intelligent adversarial attacks, its deployment in CIoT environments presents several practical challenges. Although experiments have shown that the overhead of TM-MTD is generally minimal and within an acceptable range, it is important to consider that CIoT devices are typically resource-constrained. In highly complex scenarios, this may lead to more noticeable delays, particularly for low-end devices with limited processing power.

Failure modes include the diminishing returns of frequent mutations under extreme attack scenarios. While TM-MTD improves defense adaptability, excessively high mutation frequencies can lead to less effective defense over time as the system suffers from excessive overhead. Additionally, the use of dynamic IP hopping may complicate log auditing and forensic analysis, raising concerns about traceability and transparency, which are critical in regulated environments.

Ethically and operationally, the heterogeneity of defense strategies in TM-MTD may introduce operational complexity, especially when managing diverse CIoT devices. Ensuring compatibility and efficient operation across heterogeneous devices with different capabilities can increase the administrative burden for network operators. These challenges must be carefully managed to maintain the reliability and transparency of the system while providing robust security.

IX. CONCLUSION

This article proposes a novel triple-mutation network moving target defense solution with endogenous security enhancement, designed to counter and suppress the increasingly diverse and sophisticated intelligent attacks emerging in CIoT environments. To reinforce defense systems' ability to resist exploitation without compromising performance, our hierarchical TM-MTD architecture integrates a lightweight bionic algorithm at each edge node. This self-adaptive mechanism efficiently extracts defense strategies while ensuring

minimal resource usage. Moreover, we utilizes a sub-action reinforcement learning mechanism to manage the strategic deployment across edge nodes, thereby enhancing the long-term effectiveness of the defense architecture in highly dynamic CIoT networks. Experimental results on benchmark intrusion detection datasets and IoT datasets also highlight that the TM-MTD solution can significantly enhance the resistance against multiple intelligent adversarial attacks.

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